In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE(Main) - 2017

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST - IV (Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	A	B
2.	C	C	C
3.	A	B	D
4.	C	A	A
5.	B	D	B
6.	C	D	B
7.	C	B	D
8.	D	C	A
9.	C	B	A
10.	A	B	C
11.	B	C	D
12.	C	D	C
13.	D	B	A
14.	C	A	A
15.	C	B	A
16.	A	C	D
17.	D	B	D
18.	C	B	A
19.	C	A	D
20.	B	D	C
21.	A	C	B
22.	A	C	B
23.	C	A	A
24.	D	B	A
25.	B	B	B
26.	A	D	C
27.	C	B	B
28.	B	B	C
29.	A	C	D
30.	B	D	A

Physics

PART – I

SECTION – A

3. $\vec{v}.\vec{r} \geq 0$

$2u^2 - 3u\sin\theta gt + g^2 \geq 0$

$9u^2\sin^2\theta g^2 - 8u^2g^2 \geq 0 \Rightarrow \sin\theta \geq 2\sqrt{\frac{2}{3}}$

6. From 0 to t_1 emf is constant and $-ive$

From $t_1 to t_2$ emf is zero

From t_2 to $t_1 + t_2$ emf is constant and $+ive$

7. $V_{new} = \frac{\frac{\theta}{2} + cv}{c} = V + \frac{\theta}{2c}$

8. $V = \frac{1}{4\pi \in_0}.\frac{\theta}{1+\frac{1}{2}} = \frac{\theta}{6\pi \in_0}$

10. $rate \propto T^4$

11. $\frac{dr}{dt}$=constant

12. $\alpha = \frac{\omega dw}{dv} \Rightarrow at\,\theta = 1\alpha = 4\times 3 = 12\,rod/\sec^2$

16. $\frac{1}{2}.\frac{a_1 t}{2} = \frac{1}{2}a_2(t-2) \Rightarrow a_1 t = 2a_2 t = 4a_2$

20. Angle of incident u is greater than vertical angle

21. $91.3 \times V = 0.31$

22. $L^2\sqrt{2gy} = 2\pi R^2\sqrt{2gy} \Rightarrow R = \frac{L}{\sqrt{2n}}$

25. $\frac{E_1 + E_2}{R + r_1 + r_2} > \frac{E_1}{R + r_1}$

27. $\frac{hc}{\lambda} - \phi = eV_{\max}$

28. $\frac{t}{1/2}=\frac{3}{4}t$

$\Rightarrow \frac{11}{2}t-4 \Rightarrow \frac{3}{2}t=\frac{2t}{1/2}$

29. $\left[\frac{\beta}{\alpha}\right]=\left[LT^{-1}\right]$

30. $2\pi\sqrt{\frac{L}{(mg-kv)/m}}=2\pi\sqrt{\frac{L}{g}}\times 2$

$\Rightarrow \frac{1}{mg-kv}=\frac{4}{mg}$

$\Rightarrow mg-kv=mg/4$

$v=\frac{3mg}{4k}$

Chemistry

PART – II

SECTION – A

1. Down the group covalent character decrease

3. $Na\left[\overset{+I}{A}g\left(CN\right)_2\right]$

5. $EDTA^{4-}$ is Hexadentate and form optically active compounds.

9. Using MOT (paramagnetic entities are attracted towards externally applied magnetic field) find paramagnetic ions.

11. Rate of effusion $\propto \frac{1}{\sqrt{T}}$

13. Charge is not balanced in reaction except option (B)

15. Area under the curve $= -1\times 3\times 10^{-5} - \frac{1}{2}\times 3\times 3\times 10^{-5} = -7.5\times 10^{-5}\ J$

16. $P_4 + 20HNO_3 \longrightarrow 4H_3PO_4 + 20\,NO_2\uparrow + 4H_2O$

$m.moles\ of\ NaOH = 2\times m.moles\ of\ H_3PO_4 \Rightarrow m.moles\ H_3PO_4 = \frac{100\times 0.1}{2} = 5$

$m.moles\ of\ HNO_3 = \frac{20}{4}\times 5 = 25$

Volume of $HNO_3 = 25\,ml$

20. $Zn + 2NaOH \rightarrow \underset{(A)}{Na_2ZnO_2} + \underset{(X)}{H_2}$

$Na_2ZnO_2 + H_2S \rightarrow Zns + 2NaOH$

22. Zn^{2+}, Li^+, K^+, Al^{3+} are colourless where as Co^{2+} is coloured.

27. [700]

$X_x = 0.30$

& $Y_x = 0.60$

$P_B^0 = 700\,mm$

Mathematics

PART – III

SECTION – A

1. $$\int \frac{3x^2+4x-1}{\left(x^2+1\right)^2\sqrt{x+1}}dx$$

$$=\int \frac{4x^2+4x-x^2-1}{\left(x^2+1\right)^2\sqrt{x+1}}dx$$

$$=4\int \frac{x\left(x+1\right)dx}{\left(x^2+1\right)^2\sqrt{x+1}}-\int \frac{dx}{\left(x^2+1\right)\sqrt{x+1}}$$

$$4\int \frac{x\left(x+1\right)dx}{\left(x^2+1\right)^2\sqrt{x+1}}.\left[\frac{2\sqrt{x+1}}{x^2+1}\int \frac{-1\quad 2x.2\sqrt{x+1}}{\left(x^2+1\right)^2}.dx\right]$$

$$=4\int \frac{x\left(x+1\right)dx}{\left(x^2+1\right)^2\sqrt{x+1}}-\frac{2\sqrt{x+1}}{x^2+1}$$

$$=\frac{-2\sqrt{x+1}}{x^2+1}+c$$

2. $$y\left(xy-\left(x^2-y^2\right)^2\right).dx=\left\{y^3-x\left(x^2-y^2\right)^2\right\}dy$$

$$\Rightarrow xy^2dx-y\left(x^2-y^2\right)^2dx=y^3dy-x\left(x^2-y^2\right)^2dy$$

$$\Rightarrow y^2\left(xdx-ydy\right)=\left(x^2-y^2\right)^2\left(ydx-xdy\right)$$

$$\Rightarrow xdx-ydy=\left(x^2-y^2\right)^2\frac{\left(ydx-xdy\right)}{y^2}$$

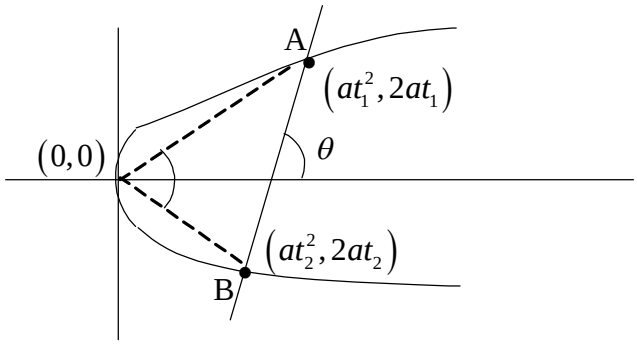
$$\frac{2\left(xdx-ydy\right)}{\left(x^2-y^2\right)^2}=\frac{2\left(ydx-xdy\right)}{y^2}$$

$$\int \frac{d\left(x^2-y^2\right)}{\left(x^2-y^2\right)^2}=2\int d\left(\frac{x}{y}\right)$$

$$\frac{-1}{x^2-y^2}=\frac{2x}{y}+C$$

$$\therefore \frac{2x}{y}+\frac{1}{x^2-y^2}=C$$

3.



$y^2 = 8x$

Now, $t_1 t_2 = -4$

And $t_2 = -t_1 - \frac{2}{t_1}$

$\frac{-4}{t_1} = -t_1 - \frac{2}{t_1}$

$\Rightarrow -t_1^2 = -2$

$\Rightarrow t_1^2 = 2$

$y^2 = 4ax$

$\frac{dy}{dx} = \frac{4a}{2y}$

$\frac{dy}{dx} = \frac{1}{t}$

Slope of normal $= -t$

And $\tan^2\theta = t_1^2 = 2$

4. $C : \left(x^2y^2 - 4x - 8y - 5\right) + \lambda\left(x^2 + y^2 - 12x + 2y + 29\right) = 0$

Centre : $\left(\frac{2+6\lambda}{1+\lambda}, \frac{4-\lambda}{1+\lambda}\right)$

Common chord $4x - 5y - 17 = 0$

Centre lies on common chord

$\therefore \lambda = \frac{29}{12}$

$\therefore$ centre $\left(\frac{1+6(29/12)}{1+29/12}, \frac{4-29/12}{1+29/12}\right)$

$\therefore$ centre $\left(\frac{198}{41}, \frac{19}{41}\right)$

5. $\int_0^{\frac{11\pi}{3}} \sqrt{1+\cos 2x}.dx$

$$= \int_0^{\frac{11\pi}{3}} \sqrt{2}\left|\cos x\right|.dx$$

$$= \int_0^{3\pi} \left|\sqrt{2}\cos x\right|.dx + \int_{3\pi}^{3\pi+2\pi/3} \sqrt{2}\left|\cos x\right| dx$$

$$3\sqrt{2}\int_0^{\pi} \left|\cos x\right|.dx + \sqrt{2}\int_0^{2\pi/3} \left|\cos x\right|.dx$$

$$= 6\sqrt{2}\int_0^{\pi/2} \left(\cos x\right) dx + \sqrt{2}\int_0^{\pi/2} \cos x.dx - \sqrt{2}\int_{\pi/2}^{2\pi/3} \cos x.dx$$

$$= 6\sqrt{2} + \sqrt{2}\frac{-\sqrt{3}}{\sqrt{2}} + \sqrt{2}$$

$$= \sqrt{2} - \sqrt{\frac{3}{2}}$$

6. $2\vec{a} - \vec{b} = \hat{i} + 3\hat{j} - 5\hat{k}$

$\vec{a} - 2\vec{b} = -4\hat{i} + 4\hat{j} - 4\hat{k}$

Area of $= \frac{1}{2}\left|\left(2\vec{a} - \vec{b}\right) \times \left(\vec{a} - 2\vec{b}\right)\right|$

$\left(2\vec{a} - \vec{b}\right) \times \left(\vec{a} - 2\vec{b}\right) = 3\hat{i} + 24\hat{j} + 15\hat{k}$

Area $= \frac{9\sqrt{10}}{2}$

7. Taking first two

$$\frac{\tan\left(\pi/4 + \alpha\right)}{\tan\left(\pi/4 + \beta\right)} = \frac{5}{3}$$

$$\therefore \frac{\cos\left(\alpha + \beta\right)}{\sin\left(\alpha - \beta\right)} = \frac{8}{2} = 4$$

$$\therefore \cos\left(\alpha + \beta\right) = 4\sin\left(\alpha - \beta\right) - - - - - - - \left(1\right)$$

Similarly

$$\cos\left(\beta + \alpha\right) = \sin\left(\beta - \alpha\right) - - - - - - \left(2\right)$$

$$3\cos\left(\gamma + \alpha\right) = -7\sin\left(\gamma - \alpha\right) - - - - - \left(3\right)$$

After marking $\sin^{-2}$ terms

$$\left(1\right) + \left(2\right) - \left(3\right)$$

Then,

$$12\sin^2\left(\alpha - \beta\right) + 15\sin^2\left(\beta - \gamma\right) - 7\sin\left(\gamma - \alpha\right) = 0$$

8. At $x = 0, y = 1$ for both curves intersect

Now $y = \left(1 + x\right)^{\cos x} + \sin x$

$$\therefore \frac{dy}{dx} = 2 \; at \; x = 0$$

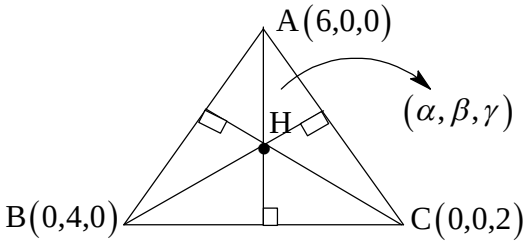
And $y=\frac{1}{2}\left(x^2+x+2\right)$

$\frac{dy}{dx}=\frac{1}{2}\ at\ x=0$

Now $m_1=2,\ m_2=\frac{1}{2}$

$\therefore \tan\theta=\left|\frac{2-1/2}{1+(2)(1/2)}\right|=\left|\frac{3/2}{2}\right|\Rightarrow\frac{3}{4}$

9.



Now $AH\perp BC$

Now D^r of $AH=(\alpha-6,\beta,\gamma)$

$D^r\ of\ BC=(0,4,-2)$

Now for 1 lines,

$a_1a_2+b_1b_2+c_1c_2=0$

$\therefore 0(\alpha-6)+\beta(4)-2\gamma=0$

$\therefore 4\beta-2\gamma=0$

$2\beta=\gamma---(1)$

Similarly

$-6\alpha+2\gamma=0-----(2)$

$-6\alpha+4\beta=0-----(3)$

$\therefore 2\beta=\gamma;\ \frac{2\beta}{3}=\alpha$

Now may lie on

$2x+3y+6z=12$

$\therefore 2\alpha+3\beta+6\gamma=12$

$2\left(\frac{2\beta}{3}\right)+3\beta+6(2\beta)=12$

$4\beta+9\beta+36\beta=36$

$49\beta=36$

$\beta=\frac{36}{49}$

Now, $\alpha=\frac{24}{49}$ and $\gamma=\frac{72}{49}$

$$(\alpha,\beta,\gamma)=\left(\frac{24}{49},\frac{36}{49},\frac{72}{49}\right)$$

10. $f(x)=\log_3\left(\log_{1/2}\left(\log_2\left(\log_{1/3}\right)\right)\right)$

$$\frac{x+1}{\sqrt{1-3x-4x^2}}$$

Now, $\sqrt{1-3x-4x^2}>0$

$4x^2+3x-1<0$

$\therefore (4x-1)(x+1)<0$

$\therefore x\in(-1,1/4)$

$\log_{1/2}\left(\log_2\left(\log_{1/3}(x)\right)\right)>0$

$\log_2\left(\log_{1/3}(x)\right)<1$

$\log_{1/3}(x)<2$

$x>1/9$

Now $F\left(f(x)\right)\in x>\frac{1}{9}\cap x\in(-1,1/4)$

$\therefore x\in\left(\frac{1}{9},\frac{1}{4}\right)$

11. $f(x)+f(1-x)=\frac{1}{27}$

$\frac{54}{27}=2$

12. $\cot\sum_{n=1}^{\infty}\cot^{-1}\frac{1}{n^2}\times n^2\frac{(n+1)^2}{2}$

$=\cot\sum_{n=1}^{\infty}\tan^{-1}\frac{2}{(n+1)^2}$

$=\cot\sum_{h=1}^{\infty}\left(\tan^{-1}(n+2)-\tan^{-1}n\right)$

$$s_n=\begin{bmatrix}\tan^{-1}3-\tan^{-1}1\\ \tan^{-1}4-\tan^{-1}2\\ \tan^{-1}(n+2)-\tan^{-1}n\end{bmatrix}$$

$s_n=\tan^{-1}\left(\frac{3n^2+7n}{n^2+9n+10}\right)$

$s_\infty=\tan^{-13}$

$$\cot\tan^{-13} = \cot\cot^{-1}\frac{1}{3} \Rightarrow \frac{1}{3}$$

13. $$\lim_{x\to\pi} a\frac{\left\{\frac{1}{1+9f(x)}\right\}9f'(x)-\cos(f(x))f'(x)}{2f'(x)}$$

$$= \frac{9-1}{2} = 4$$

$$\sum_{h=1}^{\infty} 4^{-n} = \frac{1}{4}+\frac{1}{4^2}+.....\infty$$

$$= \frac{1}{3}$$

14. Given $\frac{f(x)f'(x)}{\sqrt{1-(f(x))^4}} \geq x$

$$\frac{2f(x)f'(x)}{\sqrt{1-(f(x))^4}} - 2x \geq 0$$

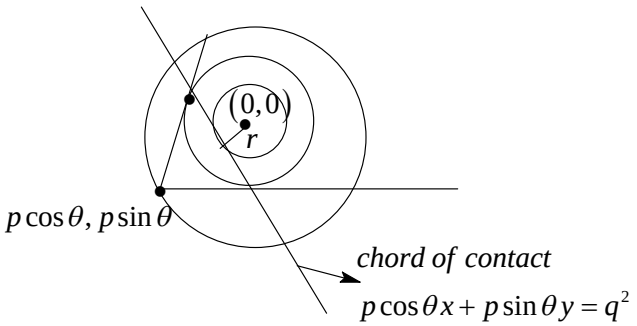
Let $g(x) = \sin^{-1}(f(x))^2 - x^2$ is non decreasing function

$$\lim_{x\to a^-} g(x) \leq \lim_{x\to b^+} g(x)$$

$$\frac{\pi}{2} - a^2 \leq \frac{\pi}{6} - b^2$$

$$a^2 - b^2 \geq \frac{\pi}{3}; \left[a^2 - b^2\right] = 1$$

15.



$\perp$ distance from (centre) to chord of contact

$$r = \left|\frac{p\cos\theta\times 0 + p\sin\theta\times 0 - q^2}{p}\right|$$

$$q^2 = rp$$

$$p, q, r \text{ are in } GP$$

$$\therefore 2a - \log_e^p,\ 4a - \log_e^q,\ 6a - \log_e^r \text{ are in } AP$$

16. *G.E.of line*

$Ax+By+C=0$

$At_1+B\left(2at_1+at_1^3\right)+c=0$

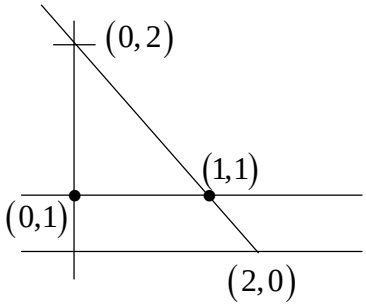
$Bat_1^3+\left(2aB+A\right)t_1+c=0$

Sum of roots

$t_1+t_2+t_3=0$

$2^{t_1}.2^{t_2}.2^{t_3}=2^\circ=1$

17.



$x^2-y^2+2y-1=0$

$x^2-\left(y-1\right)^2=0$

$\left(x+y-1\right)\left(x-y+1\right)=0$

Angle bisector

$\frac{x+y-1}{\sqrt{2}}=\pm\frac{x-y+1}{\sqrt{2}}$

$x+y-1=x-y+1$

$x+y-1=-x+y-1$

$\Rightarrow y=1 \quad \Rightarrow x=0$

Area $a=\frac{1}{2}\times1\times1=\frac{1}{2}$

Orthocentre $\left(0,1\right)\Rightarrow b=0,c=1$

$\therefore 4a+b+c=2+0+1=3$

18. $z^5=1$

$\frac{z^5}{z^5+z}+\frac{z}{z+1}+\frac{z^2}{z^2+1}+\frac{z^5}{z^2+z^5}$

$=\frac{1}{1+z}+\frac{z}{1+z}+\frac{z^2}{z^2+1}+\frac{1}{z^2+1}$

$=1+1=2$

19. Equation of tangent $y=3x\pm\sqrt{9a^2+b^2}$

This is normal to circle

$x^2+y^2+4x+1=0$

Panes – 1 brought $(-2,0)$

$\Rightarrow 0=-6\pm\sqrt{9a^2+b^2}$

$36=9a^2+b^2$

$AM \geq GM$

$\frac{9a^2+b^2}{2} \geq \sqrt{9a^2b^2}$

$18 \geq 3ab$

$6 \geq ab$

20. $F(x)=\int_0^x f(t)dt$

$F(x^2)=\int_0^{x^2} f(t)dt$

$=x^2(1+x)$

Diff. both side

$F'(x^2).2x=2x+3x^2$

$f(x^2)=1+\frac{3}{2}x$

$x=2,\ f(4)=1+\frac{3x}{2}$

$x=2,\ f(4)=1+\frac{3}{2}\times 2=4$

Unit place $(985697)^{16}=1$

26. $A=x^3-\frac{1}{x^3}=\left(x-\frac{1}{x}\right)^3+3\left(x-\frac{1}{x}\right)$

$A=B^3+3B$

$\frac{A}{B^2}=B+\frac{3}{B}$

Let $t=x-\frac{1}{x}$

$\frac{A}{B^2}=s=t+\frac{3}{t}$

$\frac{ds}{dt}=1-\frac{3}{t^2}$

$\frac{d^2s}{dt^2}=\frac{6}{t^3}$

$\frac{ds}{dt}=0 \Rightarrow t=\pm\sqrt{3}$

For min $t = +2\sqrt{3}$

$s = \sqrt{3} + \dfrac{3}{\sqrt{3}} = 2\sqrt{3}$

27. $f(x) + f(x+2) = \sqrt{3} f(x+1)$

$x \to x+2 ------(1)$

$f(x+2) + f(x-14) = \sqrt{3} f(x+3) ------(2)$

$(1)+(2)$

$f(x) + 2 + (x+2) + f(x+14)$

$= \sqrt{3}\{ f(x+1) + f(x+3)\}$

$= 3 f(x+2)$

$f(x) + f(x+4) = f(x+2)$

$x \to x+2 -----(3)$

$f(x+2) + f(x-16) = f(x+4) ---(4)$

$(3)+(4)$

$f(x) + f(x+6) = 0$

$f(x+6) = -f(x)$

$f(x+12) = -f(x+6)$

$= f(x)$

Period $T = 12$

$\dfrac{T}{2} = 6$

28. Centre $(0, -f)$

Equation of circle

$x^2 + y^2 + 2fy + c = 0$

$\left(\log_e a_r\right)^2 + \dfrac{1}{\left(\log_e a_r\right)^2} + 2f\left(\dfrac{1}{\log_e a_r}\right) + c = 0$

$\left(\log a_r\right)^4 + c\left(\log a_r\right)^2 + 2f\left(\log a_r\right) = 0$

$\log a_1, \log a_2, \log a_3, \log a_4$ are roots of equation

$\therefore \log\left(a_1.a_2.a_3.a_4\right) = 0$

$a_1 a_2 a_3 a_4 = 1$

$\dfrac{a_1 + a_2 + a_3 + a_4}{4} \geq (1)^{1/4}$

$\therefore a_1 + a_2 + a_3 + a_4 \geq 4$

29. $20c_0 \times 30c_{5m} + 20c_1 \times 30c_{5m-1}$

$+.....+ 20c_{5m} \times 30c_0$

Coeff of $x^{5m} \to (1+x)^{20}(1+x)^{30}$

$50c_{5m}, 5m = \frac{50}{2} = 25$

$m = 5$

30. $1920 = 5\times3\times2^7 = 5\times3\times2\times2\times2\times2\times2\times2\times2 = \frac{9!}{7!} = 9\times8$

$1920 = 5\times3\times4\times2\times2\times2\times2\times2\times1 = \frac{9!}{5!} = 9\times8\times7\times6$

$1920 = 3,5,4,4,2,2,2,1,1$

$\frac{9!}{2!2!3!}$

$3,5,4,4,4,2,1,1,1$

$\frac{9!}{3!3!}$

Total = $\frac{9!}{7!} + \frac{9!}{5!} + \frac{9!}{2!2!3!} + \frac{9!}{3!3!}$

$= \frac{9!}{7!}(1+42+210+140)$

$=393\ {}^9P_2$